

CAPITAL ONE · BANKING

A regulated bank that ships many times a day

How Capital One moved from monthly releases to many a day, and exited its data centres, by making automation the control. Plus how one product team can apply the same pattern in weeks.

3 mo → mins

to build a dev environment

-50%

critical incident resolution time

8

data centres exited

Industry	Banking and financial services (regulated)
Problem	Slow, manual release and provisioning in a heavily regulated environment
Approach	DevOps plus cloud, with security and compliance built into the platform
Source	AWS case study: Capital One [1]

01 The challenge

Regulated businesses often assume that control means slowness: manual approvals, change boards and long provisioning queues. At Capital One, application releases happened monthly or quarterly, and standing up a development environment could take months [1].

02 What Capital One did

- **Automation as the control.** Security and compliance checks were built into the platform and pipeline, so they ran on every change instead of in a review queue.
- **Rebuilt for the cloud.** Around 80% of its nearly 2,000 applications were rebuilt from the ground up to run natively in the cloud [1].
- **Self-service environments.** Engineers could provision a development environment in minutes rather than waiting months.
- **A long-term commitment.** The migration ran over eight years and ended with all eight on-premises data centres closed [1].

03 The results

- Releases went from monthly or quarterly to multiple times a day [1].
- Critical incident resolution time and transaction errors were both cut by 50% [1].
- Development environments that took three months now take minutes [1].
- Disaster recovery improved by 70% in testing exercises [1].

The lesson: in a regulated business, automation doesn't weaken control, it proves it. A pipeline that records every check is a better audit trail than a spreadsheet of approvals.

04 At your scale: a 90-day plan

You aren't rebuilding 2,000 applications over eight years. One product team can adopt the same pattern in a quarter.

Days 1–30 · Gap assessment	Days 31–60 · Guardrails as code	Days 61–90 · Evidence by default
• Map the controls you're assessed on (SOC 2, ISO 27001 or a customer questionnaire) • Inventory access, secrets and environments • Rank gaps by audit risk	• A landing zone built as code, with separate accounts per environment • Least-privilege access and single sign-on • Secrets moved into a managed vault; CIS-hardened baselines	• Dependency, container and code scanning in every pipeline • Approvals and change history recorded automatically • An evidence pack you can hand to auditors

05 Readiness checklist

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|--|--|
| ☐ No secrets live in code or chat | ☐ Vulnerability scans run on every build |
| ☐ Access is reviewed and least-privilege | ☐ Environments can be rebuilt from code |
| ☐ Production changes only go through the pipeline | ☐ Logs are centralised and retained |
| ☐ Every change has a recorded approval | ☐ You can produce audit evidence in a day |

Want this plan for your team?

A 30-minute call: we'll map your audit gaps and tailor the 90 days.

[Book a discovery call](#)

Sources

1. AWS, "Capital One Completes Migration from Data Centers to AWS": aws.amazon.com/solutions/case-studies/capital-one-all-in-on-aws
2. SiliconANGLE, "Capital One closes all data centers" (December 2020): siliconangle.com

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